

National Bank Financial Reporting and Supervision System

Programming with .NET

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1 Introduction

Background

The banking and financial sector play a crucial role in maintaining economic stability and public trust. Central national banks are responsible for supervising commercial banks, ensuring compliance with regulations, and monitoring financial health through periodic reports. Traditionally, many of these processes are handled through fragmented systems, email-based communications, and even mutual document submissions, which increases the risk of errors, delays, and a lack of transparency.

With the rapid advancement of information systems and regulatory technology, there is a growing demand for centralized, secure, and auditable platforms that enable efficient communication between commercial banks and supervisory authorities. Web-based information systems built with modern frameworks enable regulatory bodies to enforce standardized workflows, ensure accountability, and maintain historical records for audit and analysis purposes.

Motivation

The motivation behind this project is to design and implement a centralized web-based system that simplifies and standardizes the interaction between commercial banks and the national bank. By automating bank registration, financial report submission, and approval workflows, the system reduces administrative overhead and minimizes human error. From a technical perspective, the project also serves as an opportunity to apply core concepts of ASP.NET Core Identity in a real-world application.

Problem Statement

The main problem addressed by this project is the lack of an integrated system for managing bank registration and financial reporting workflows. Existing manual or semi-digital processes suffer from several issues, including: Lack of centralized oversight, inconsistent data formats, limited traceability and auditability, and last but not least, delayed approval and feedback.

The goal of this system is to provide a secure, role-based platform that enables structured data submission, review, and decision-making whilst maintaining a clear audit trail.

2 Related Work

Several financial reporting and banking supervision systems exist globally, often implemented as enterprise-scale solutions tailored to specific regulatory frameworks. These systems typically provide data collection, validation, and reporting capabilities, but are expensive, complex, and not easily adaptable for education or small-scale environments.
Compared to existing solutions, the proposed system focuses on simplicity, clarity, and educational value while still reflecting real-world workflows. Its novelties lie in: Clear separation of user roles (i.e., Commercial Banks and National Bank Administrator), Explicit approval workflows for both bank registration and financial reports, integration of authentication and authorization using ASP.NET Core Identity, and the use of MVC and View models to enforce separation of concerns.
A system fit both areas as a learning project and a conceptual prototype for future regulatory reporting platforms.

3 Implementation

This project was implemented using modern Microsoft development tools and frameworks selected for their robustness, scalability, performance, and industry relevance.

3.1 Tools & Technology

This project was implemented using modern Microsoft development tools and frameworks selected for their robustness, scalability, performance, and industry relevance.

3.1.1 SQL Management Studio

SQL Server Management Studio was used to manage and inspect the relational database. It allows developers to view tables, relationships, constraints, and data stored by the application. SSMS also supports query execution and database maintenance, which is essential for testing and development.

3.1.2 Visual Studio

Visual Studio was used as the primary Integrated Development Environment. It provides advanced debugging tools, project templates for ASP.NET Core MVC integrated with NuGet package management, and seamless support for Entity Framework Core and Identity Scaffolding.

Microsoft.AspNetCore.Identity.EntityFrameworkCore	Manages user authentication and roles
Microsoft.EntityFrameworkCore.SqlServer	Provides the database provider for SQL Server
Microsoft.EntityFrameworkCore.Tools	Enables migrations and database updates via the CLI

3.1.3 .NET 9.0

.NET 9.0 was utilized as the core framework, providing enhanced performance for middleware, improved security for Identity, and seamless integration with Entity Framework Core. This version allows for optimized AOT (Ahead-of-Time) compilation and refined JSON serialization, which are critical for processing high volumes of financial data.

3.1.4 ASP.NET Core MVC & Entity Framework Core

ASP.NET Core MVC was used to structure the application using the Model-View-Controller pattern. Entity Framework Core was employed as the Object-Relational Mapper to handle the database access and persistence while abstracting low-level SQL operations.

4. Database Description

The database layer represents the core persistence and integrity mechanism of the National Bank Financial Reporting and Supervision System. It is responsible for storing regulatory data, enforcing structural constraints, maintaining secure user access, and preserving a complete historical record of financial submissions and supervisory actions. The database is implemented using Microsoft SQL Server and is accessed exclusively through Entity Framework Core using the Code-First approach. This design ensures strong consistency between the object-oriented domain model and the relational schema while enabling automated schema evolution through controlled migrations.

From an architectural perspective, the database is designed to reflect real-world regulatory interactions rather than abstract technical constructs. At a conceptual level, it models the relationship between commercial banks that submit financial data, users who act on behalf of those institutions or the national bank, and the supervisory processes that govern registration, reporting, review, and approval. Surrounding these core domain entities is a comprehensive identity and security infrastructure provided by ASP.NET Core Identity, which enforces authentication, authorization, and role-based access control.

The use of a relational database is particularly suitable for this system due to the structured nature of financial data, the need for transactional integrity, and the importance of referential consistency between banks, users, and reports. SQL Server provides ACID-compliant transactions, robust indexing mechanisms, and mature security features, all of which are critical in a regulatory environment where data accuracy and traceability are mandatory.

4.1.1 Conceptual Overview of stored data

The database stores four primary categories of data. The first category includes institutional data describing commercial banks, such as identification information, approval status, and registration metadata. The second category consists of user-related data, which defines who can access the system, under which role, and on behalf of which institution. The third category stores financial and regulatory data, including periodic financial reports and regulatory circulars issued by the national bank. The fourth category includes audit and infrastructure data, which ensures accountability, traceability, and schema version control.
This layered data organization allows the system to enforce clear boundaries between business logic, security concerns, and operational metadata while still enabling efficient querying and reporting across these layers.

4.1.2 Figure and tables

Figure 1 illustrates the complete ER diagram of the system. It visualizes the relationships between banks, users, financial reports, regulatory circulars, and audit logs, as well as the Identity-related tables that enforce role-based access control. The diagram highlights one-to-many relationships, optional associations, and identity-managed join tables that collectively ensure both functional correctness and regulatory traceability.

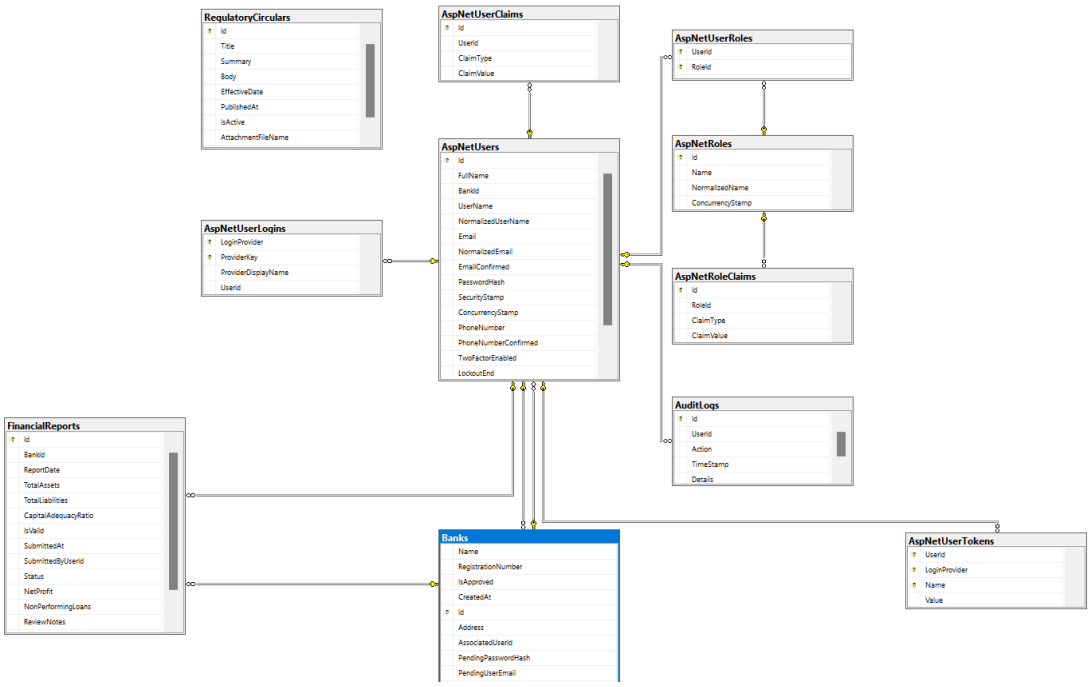


Figure 1: Complete ER Diagram of the system

While the physical implementation of the database includes several auto-generated tables required by the ASP.NET Core Identity framework—such as those for managing user claims, logins, and security tokens—this project focuses on the core business domain. Consequently, our ER analysis prioritizes five primary entities: Banks, ApplicationUsers, FinancialReports, AuditLogs, and RegulatoryCirculars. These tables represent the functional heart of the supervision system, defining the specific interactions between commercial entities and the National Bank.

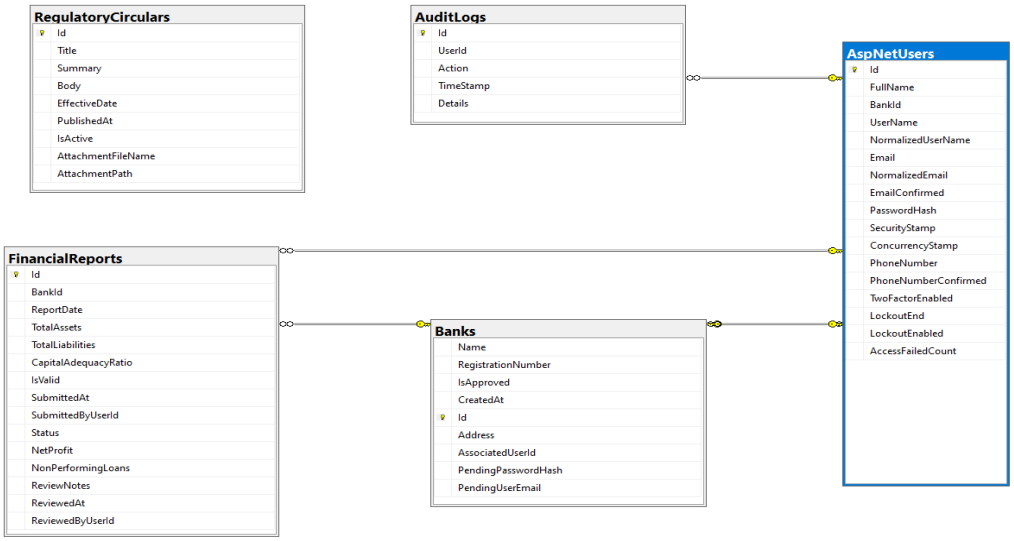


Figure 2: Primary entity models and their relationships

4.2 Database ER Diagram

4.2.1 ApplicationUser and Identity Data

All authenticated users of the system are represented by the ApplicationUser entity, which is physically stored in the AspNetUsers table generated by ASP.NET Core Identity. This table serves as the central authentication authority for the application and contains all essential credentials and security-related information required to manage user access securely.

Id	FullName	BankId	UserName	NormalizedUse...	Email	NormalizedEm...	EmailConfirmed	PasswordHash	SecurityStamp	ConcurrencySt...
bc9-0f9b92af6fee	Summit Nation...	*NULL*	finance@summ...	FINANCE@SU...	finance@summ...	FINANCE@SU...	True	AQAAAAIAAYa...	PVCYGNVYAJE5...	638c2410-f5cf-...
45f2b42c-8ad7...	Capital Bank Pri...	*NULL*	capitalbank@g...	CAPITALBANK...	capitalbank@g...	CAPITALBANK...	True	AQAAAAIAAYa...	KNRHUVYYGW...	abe32166-63e...
5b46486d-995...	Metropolitan C...	*NULL*	reports@metro...	REPORTS@MET...	reports@metro...	REPORTS@MET...	True	AQAAAAIAAYa...	XZVAFUCDSSA...	5a385986-9ab...
7967a46d-5c5f...	Heritage Saving...	*NULL*	submissions@h...	SUBMISSIONS...	submissions@h...	SUBMISSIONS...	True	AQAAAAIAAYa...	YK3BD7XWDG...	6633754f-dd3a...
ade0b63e-6ef9...	National Bank ...	*NULL*	admin@nation...	ADMIN@NATI...	admin@nation...	ADMIN@NATI...	True	AQAAAAIAAYa...	IMJXDZJWDUC...	d65bde0e-1be...
c0cdcbdb-ef67...	Riverside Com...	*NULL*	reports@riversi...	REPORTS@RIV...	reports@riversi...	REPORTS@RIV...	True	AQAAAAIAAYa...	EEP5W6ALZZ2...	142c74f9-3858...
c9877b46-fcac-...	Continental Inv...	*NULL*	reporting@con...	REPORTING@C...	reporting@con...	REPORTING@C...	True	AQAAAAIAAYa...	25Z23ZEEURQ...	3fa8f52a-8599-...
ca44aa2a-2806...	Pioneer Comm...	*NULL*	finance@pione...	FINANCE@PIO...	finance@pione...	FINANCE@PIO...	True	AQAAAAIAAYa...	DGXUJMNPXH...	27121d57-c91...
dd693cfa-21f2-...	The Global Rese...	*NULL*	reporting@glo...	REPORTING@G...	reporting@glo...	REPORTING@G...	True	AQAAAAIAAYa...	FT2S4GNLCZDI...	1eee1e6d-6262...
e4c3a81b-badf...	Pacific Rim Fina...	*NULL*	finance@pacific...	FINANCE@PAC...	finance@pacific...	FINANCE@PAC...	True	AQAAAAIAAYa...	CR4TQAZTMSA...	06f32c16-885d...
e9c6ea40-d274...	S-Bank Primary ...	*NULL*	sbank@gmail.c...	SBANK@GMAIL...	sbank@gmail.c...	SBANK@GMAIL...	True	AQAAAAIAAYa...	WRNC6IWUP3...	3ed87815-db3...
ffc9498d-ee91-...	SKBank Primary...	*NULL*	skbank@gmail....	SKBANK@GMA...	skbank@gmail....	SKBANK@GMA...	True	AQAAAAIAAYa...	YQENYCKLITEIU...	a1cd6e87-03bf...

Figure 3: ApplicationUser table and its contents

In addition to the default fields provided by the Identity framework, such as username, email, password hash, security stamp, and lockout indicators. The user model is extended with custom domain-specific attributes. The inclusion of a FullName field improves traceability and administrative clarity, while the optional BankId foreign key establishes a logical association between commercial bank users and the institutions they represent. This design allows the system to distinguish clearly between national bank administrators, who operate independently of any commercial entity, and commercial bank users, whose permissions and actions are institution-bound.

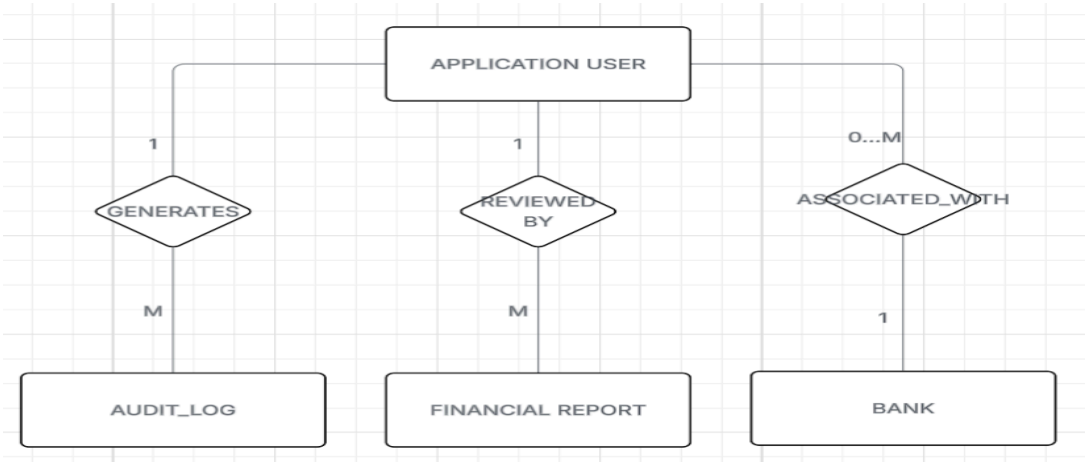


Figure 4: ApplicationUser and its relationship with other entities

The Identity subsystem further maintains several supporting tables that work in conjunction with AspNetUsers. Role assignments are managed through a many-to-many relationship implemented via the AspNetUserRoles join table, linking users to predefined roles such as National Bank Administrator and Commercial Bank User. Additional tables handle claims, external logins, and authentication tokens, enabling extensible authorization policies and modern security practices without requiring custom implementations.

4.2.2. Bank Entity

The Bank entity represents a registered commercial bank operating under the supervision of the national bank. This table stores essential regulatory identification data, including the bank's official name, registration number, physical address, approval status, and timestamps indicating when the institution was created and approved within the system.

Name	RegistrationNu...	IsApproved	CreatedAt	Id	Address	AssociatedUse...
The Global Rese...	GRT-2024-001	True	2025-12-05 11:...	1	123 Financial Pl...	dd693cfa-21f2-...
Pioneer Comm...	PCF-2024-002	True	2025-10-14 11:...	2	456 Main Street...	ca44aa2a-2806...
Metropolitan C...	MCB-2024-003	True	2025-09-05 11:...	3	789 Business B...	5b46486d-995...
Heritage Saving...	HST-2024-004	True	2025-11-29 11:...	4	321 Heritage La...	7967a46d-5c5f...
Pacific Rim Fina...	PRF-2024-005	True	2025-10-22 11:...	5	654 Ocean Driv...	e4c3a81b-badf...
Continental Inv...	CIB-2024-006	True	2025-09-24 11:...	6	987 Investment...	c9877b46-fcac-...
Summit Nation...	SNB-2024-007	True	2025-12-06 11:...	7	147 Summit Av...	3a8211df-d5fa...
Riverside Com...	RCB-2024-008	True	2025-12-07 11:...	8	258 Riverside D...	c0cdcbdb-ef67...
Capital Bank	BTR-123-4DD	True	2026-01-13 12:...	9	567.Financial Di...	45f2b42c-8ad7...
S-Bank	TRT-2026-21	True	2026-01-21 20:...	10	590.Finance Dis...	e9c6ea40-d274...
SKBank	SK-2026-19	True	2026-01-21 20:...	11	499 Metropolis,...	ffc9498d-ee91-...

Figure 5: Bank enity table and its contents

Each bank exists as an independent regulatory subject and may be associated with multiple users and multiple financial reports over time. The approval status field plays a central role in enforcing regulatory workflows, as only approved banks are permitted to submit financial reports. This constraint is enforced both at the application logic level and through relational integrity within the database.

The separation of bank data into its own entity ensures that institutional information remains consistent and reusable across multiple reporting periods and user accounts, preventing data duplication and improving long-term maintainability.

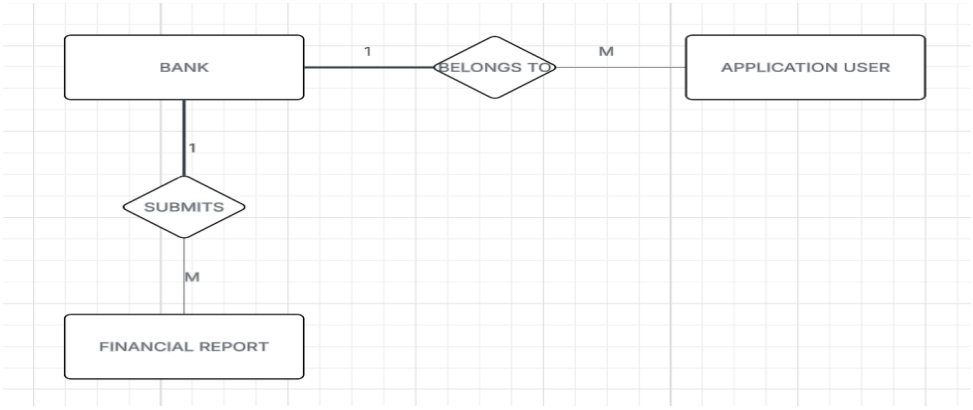


Figure 6: Bank and its relationship with other entities

4.2.3 Financial Report Entity

The FinancialReport entity stores structured financial data submitted periodically by commercial banks for regulatory review. Each report includes quantitative indicators such as total assets, total liabilities, capital adequacy ratio, net profit, and non-performing loan ratios. These values are stored using appropriate numeric data types to preserve precision and enable future analytical processing.

Id	BankId	ReportDate	TotalAssets	TotalLiabilities	CapitalAdequa...	IsValid	SubmittedAt	SubmittedByU...	Status	NetProfit	NonPerfc
1	1	2025-10-13 11:...	236406157.00	200945233.45	15.0000	True	2025-10-16 11:...	dd693cfa-21f2-...	0	1954488.00	0.8500
2	1	2025-11-13 11:...	374225596.00	329318524.48	12.0000	True	2025-11-17 11:...	dd693cfa-21f2-...	0	716834.00	1.9100
3	1	2025-12-13 11:...	295161058.00	227274014.66	23.0000	True	2025-12-16 11:...	dd693cfa-21f2-...	0	94034.00	4.3600
4	2	2025-07-13 11:...	338708215.00	257418243.40	24.0000	True	2025-07-14 11:...	ca44aa2a-2806...	1	685834.00	2.5100
5	2	2025-08-13 11:...	96180013.00	75982210.27	21.0000	True	2025-08-16 11:...	ca44aa2a-2806...	0	454109.00	4.3900
6	2	2025-09-13 11:...	160696374.00	128557099.20	20.0000	True	2025-09-17 11:...	ca44aa2a-2806...	0	1759284.00	1.5900
7	2	2025-10-13 11:...	158503252.00	118877439.00	25.0000	True	2025-10-14 11:...	ca44aa2a-2806...	1	1370087.00	0.0600
8	2	2025-11-13 11:...	98951928.00	76192984.56	23.0000	True	2025-11-14 11:...	ca44aa2a-2806...	0	-332766.00	0.6100
9	2	2025-12-13 11:...	396215488.00	348669629.44	12.0000	True	2025-12-14 11:...	ca44aa2a-2806...	0	1407676.00	0.9000
10	3	2025-09-13 11:...	372069372.00	279052029.00	25.0000	True	2025-09-14 11:...	5b46486d-995...	1	105036.00	3.7800
11	3	2025-10-13 11:...	170908835.00	135017979.65	21.0000	True	2025-10-16 11:...	5b46486d-995...	0	-130150.00	3.0000
12	3	2025-11-13 11:...	124094335.00	99275468.00	20.0000	True	2025-11-17 11:...	5b46486d-995...	1	858740.00	0.1400
13	3	2025-12-13 11:...	183715193.00	150646458.26	18.0000	True	2025-12-14 11:...	5b46486d-995...	0	312282.00	0.3400
14	4	2025-10-13 11:...	248200241.00	176222171.11	29.0000	True	2025-10-17 11:...	7967a46d-5c5f...	1	47746.00	1.1000
15	4	2025-11-13 11:...	395075170.00	331863142.80	16.0000	True	2025-11-16 11:...	7967a46d-5c5f...	0	869191.00	1.6700
16	4	2025-12-13 11:...	309725097.00	238488324.69	23.0000	True	2025-12-15 11:...	7967a46d-5c5f...	0	1673055.00	1.4300
17	5	2025-08-13 11:...	472078517.00	410708309.79	13.0000	True	2025-08-17 11:...	e4c3a81b-badf...	1	36698.00	2.8400
18	5	2025-09-13 11:...	128088226.00	111436756.62	13.0000	True	2025-09-14 11:...	e4c3a81b-badf...	0	713955.00	0.3500

Figure 7: FinancialReport table and its contents

Each financial report is linked to exactly one bank through a foreign key relationship, establishing a clear ownership and accountability model. In addition to financial metrics, the table stores regulatory metadata such as submission date, review status, reviewer comments, and reviewer identity. This information supports the full lifecycle of regulatory supervision, from initial submission to final approval or rejection.

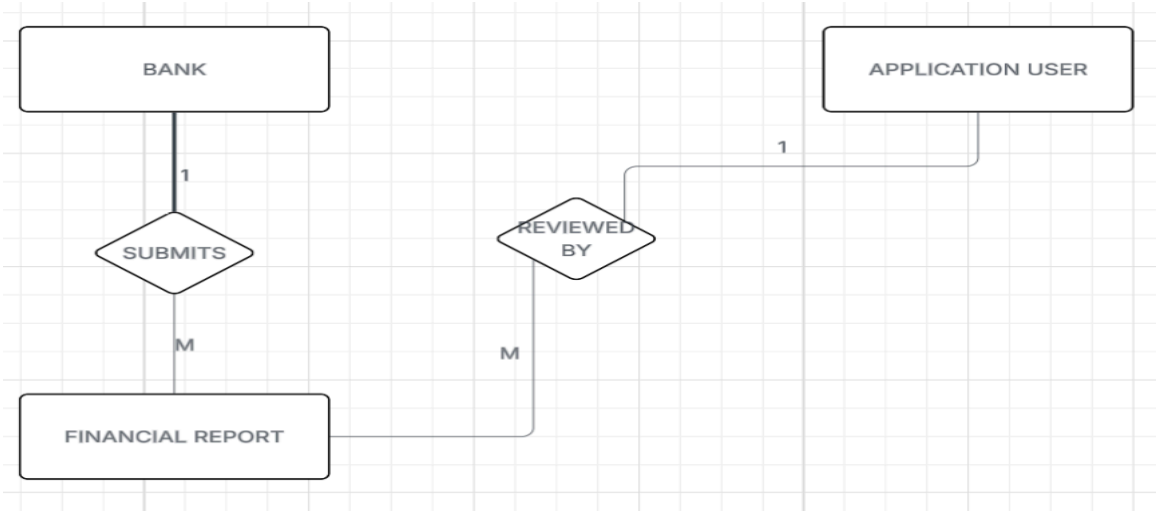


Figure 8: FinancialReport and its relationship with other entities

By retaining all submitted reports rather than overwriting previous data, the system supports historical analysis and longitudinal supervision, allowing regulators to observe trends and identify emerging risks over time.

4.2.4 Regulatory Circular and audit log entity

The RegulatoryCircular entity represents official communications issued by the national bank to commercial banks. These circulars may include regulatory updates, reporting guidelines, or compliance requirements. Storing circulars in the database ensures that all institutions have access to the same authoritative information and that communication history is preserved.

Id	Title	Summary	Body	EffectiveDate	PublishedAt	IsActive	AttachmentFile...	AttachmentPath
1	Test 1	Test 1 Summary	Test 1 Details	2026-01-20 00:...	2026-01-20 10:...	True	*NULL*	*NULL*

Figure 9: RegulatoryCircular table and its contents

The AuditLog entity provides a comprehensive record of sensitive system actions, including bank approvals, report submissions, status changes, and administrative decisions. Each audit log entry records the action performed, the user responsible, the affected entity, and the timestamp of the operation. This table plays a critical role in ensuring accountability and supporting post-incident analysis or regulatory audits.

Id	UserId	Action	TimeStamp	Details
4	45f2b42c-8ad7...	UserLogout	2026-01-15 19:...	User logged ou...
5	ade0b63e-6ef9...	UserLogin	2026-01-15 19:...	User logged in: ...
6	ade0b63e-6ef9...	UserLogout	2026-01-15 19:...	User logged ou...
7	ade0b63e-6ef9...	UserLogin	2026-01-15 19:...	User logged in: ...
8	ade0b63e-6ef9...	UserLogout	2026-01-20 10:...	User logged ou...

Figure 10: AuditLog table and its contents

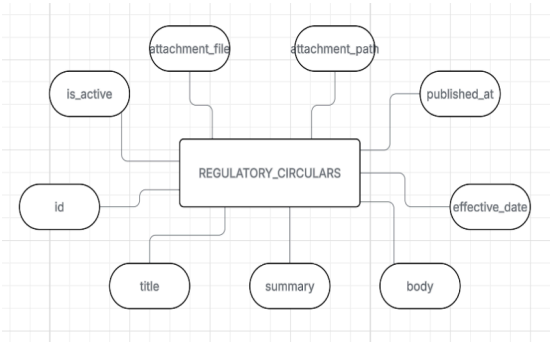


Figure 11: RegulatoryCircular entity and attributes

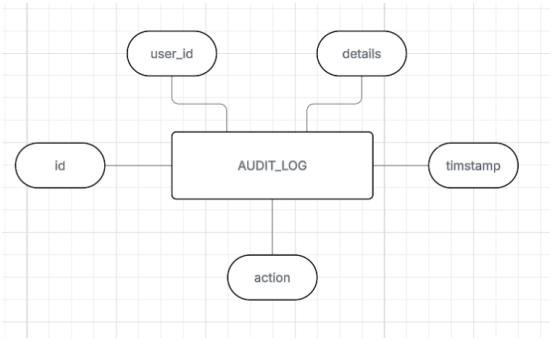


Figure 12: AuditLog entity and its attributes

4.2.5 Infrastructure and Migration tables

In addition to business and security entities, the database includes infrastructure tables automatically maintained by Entity Framework Core. The __EFMigrationsHistory table tracks applied migrations and ensures that the database schema remains synchronized with the application's domain model. This mechanism enables controlled schema evolution and simplifies deployment across development and production environments.

While these tables are not directly involved in business logic, they are essential for maintaining system stability and consistency over time.

5 System Details and Diagram

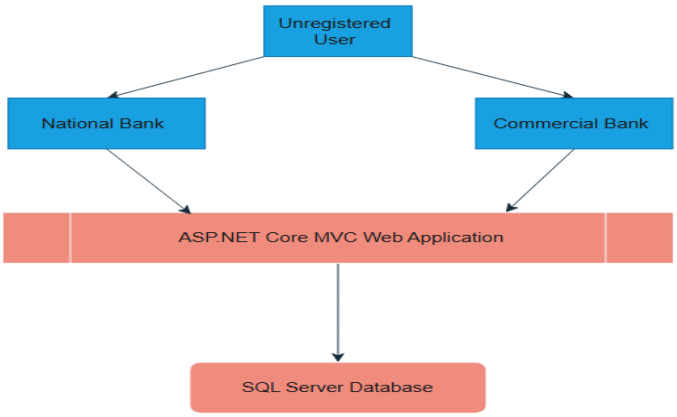


Figure 13: The overall system structure and support

The system implements a granular authorization model to ensure data privacy and regulatory integrity, supporting three primary functional roles:

Public (Unregistered) User: Represents the initial state of an entity before authentication or during the registration phase. Their primary capability is to initiate formal bank registration requests by providing institutional metadata and proposed credentials.

Commercial Bank User: A verified institutional representative granted access following administrative approval. These users are authorized to submit periodic financial reports, monitor submission history, and review regulatory feedback or rejection notes.

National Bank Administrator: The supervisory authority with comprehensive oversight capabilities. Administrators are responsible for auditing bank registrations, performing technical reviews of financial submissions, validating compliance metrics, and accessing high-level data dashboards.

So the onboarding process follows a secure “verification-first” approach:

1. Request: A bank submits details via the public portal
2. Audit: The admin verifies the bank's registration number offline
3. Provisioning: Upon approval, the system automatically creates the ApplicationUser account and assigns the Commercial Bank User role, sending a confirmation to the bank.

5.1 Core Workflow

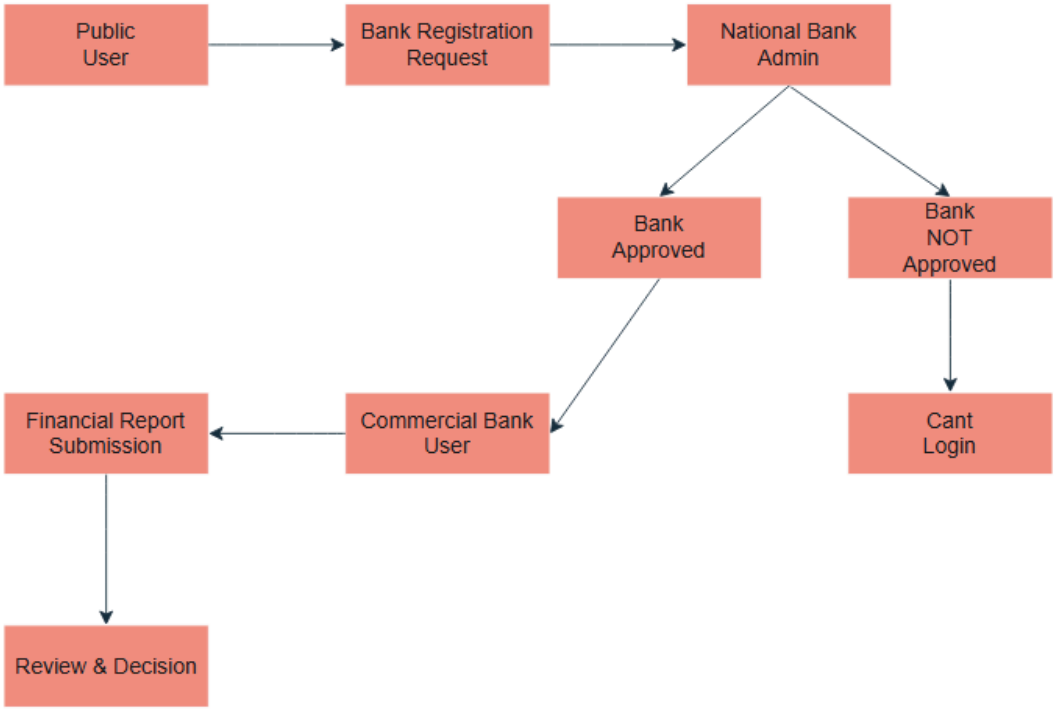


Figure 14: Public User to Commercial Bank User

The figure above portrays a structured lifecycle for financial oversight, beginning with the transition of a Public User to a verified Commercial Bank User. This onboarding process is strictly governed by the National Bank Administrator, who retains the sole authority to audit registration requests. In instances of denial, the system enforces a security block, preventing the creation of a profile and subsequent access to the workspace. Conversely, upon administrative approval, the user is granted access to a personalized dashboard to fulfill their reporting obligations. Central to this operational phase is the submission of monthly financial data, where the system employs automated logic to ensure institutional health.

During submission, the platform calculates the Capital Adequacy Ratio (CAR) by applying the formula:
CAR = ((Total Assets - Total Liabilities) / Total Assets) * 100

This computational layer acts as an immediate regulatory filter; if the resulting ratio falls below established safety thresholds, the report is automatically flagged, alerting the Administrator to prioritize the review of the potentially high-risk institution. This integration of human oversight and automated validation ensures a proactive approach to national financial supervision.

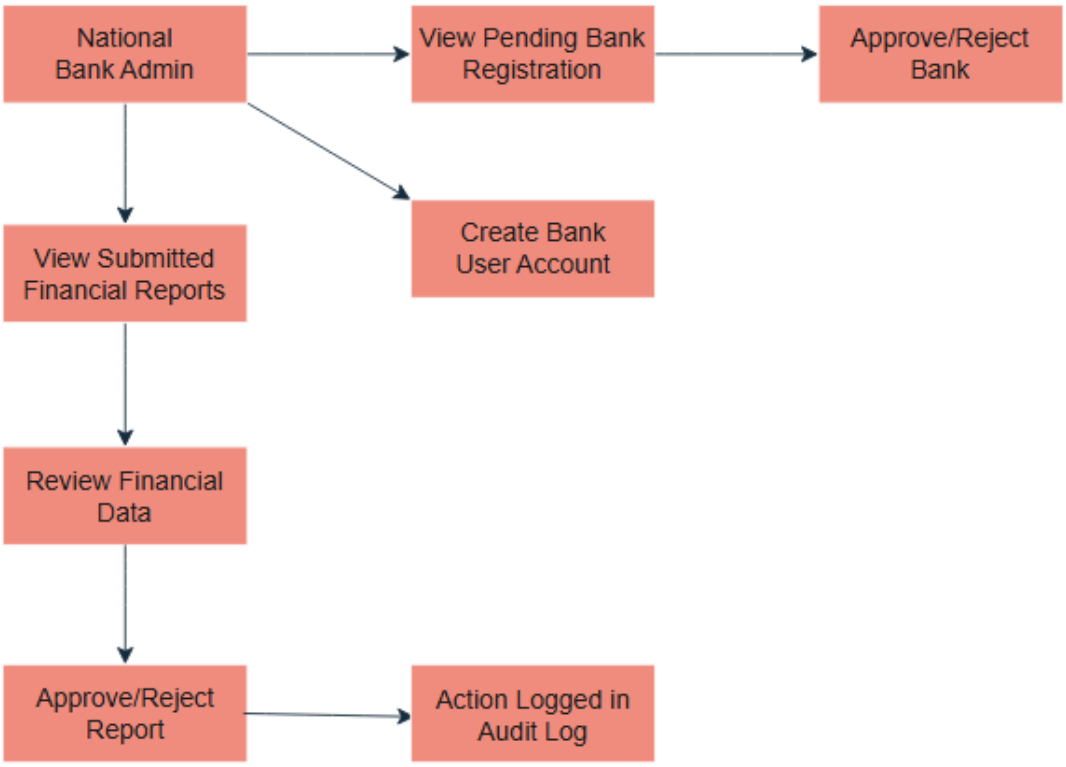


Figure 15: Admin Functions

This figure now portrays the functions and responsibilities from the perspective of the National Bank Administrator. The National Bank Administrator serves as the central authority within the system, possessing broad powers to manage institutional access and maintain financial integrity. The administrator's responsibilities are categorized into three primary regulatory functions:

Institutional Vetting and Registration Management: The administrator is responsible for auditing all pending commercial bank registration requests initiated by public users. They maintain the exclusive authority to approve or deny access based on external verification of the bank's credentials.
Dual-Path Onboarding: The system supports two distinct registration workflows to ensure flexibility. While commercial banks typically initiate requests through the public portal for subsequent review, the administrator also can perform "Direct Registration." This internal process allows the National Bank to bypass the pending state and immediately provision an active, automatically approved institutional account.
Financial Audit and Feedback Loop: The final core function involves the systematic review of submitted financial reports. The administrator examines the data for compliance, using the system's automated flags as a guide, and issues a formal decision to approve or reject the submission. This process includes the ability to provide optional qualitative feedback or review notes, ensuring that commercial banks receive clear guidance on regulatory discrepancies.

6 System Workflows and operational Processes

6.1 Public user perspective

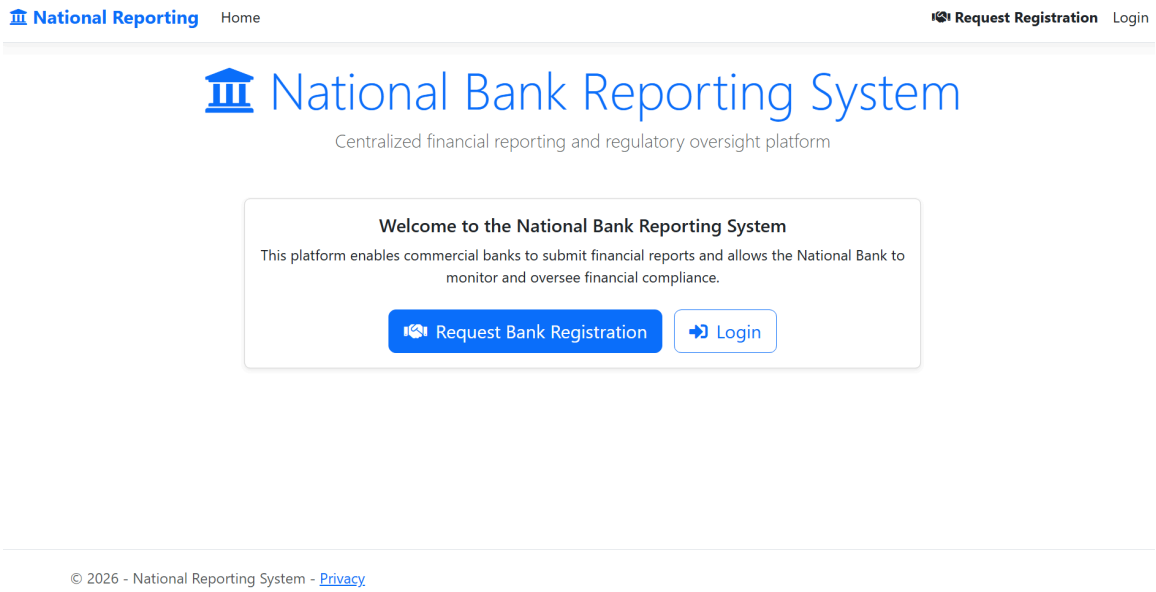


Figure 16: National Bank Reporting System Landing Page

The landing page serves as the initial gateway for all three user types (Public, Commercial, and Admin). At this stage, the system remains in a 'logged-out' state, enforcing a strict boundary between public information and sensitive regulatory data.

Workflow for Unregistered Entities:

1. Request-Only Access: Unregistered commercial banks are restricted from logging in. Their only available action is to initiate a Registration Request, providing institutional metadata that must be vetted by the National Bank.
2. Secure Onboarding: This prevents unauthorized account creation and ensures that every commercial bank in the system has been manually verified against official records.

Workflow for Registered Entities:

1. Direct Authentication: Users with existing credentials, including both previously approved Commercial Bank representatives and National Bank Administrators, can bypass the registration portal to access their respective secure dashboards via the Login portal."

6.2 Administrator perspective

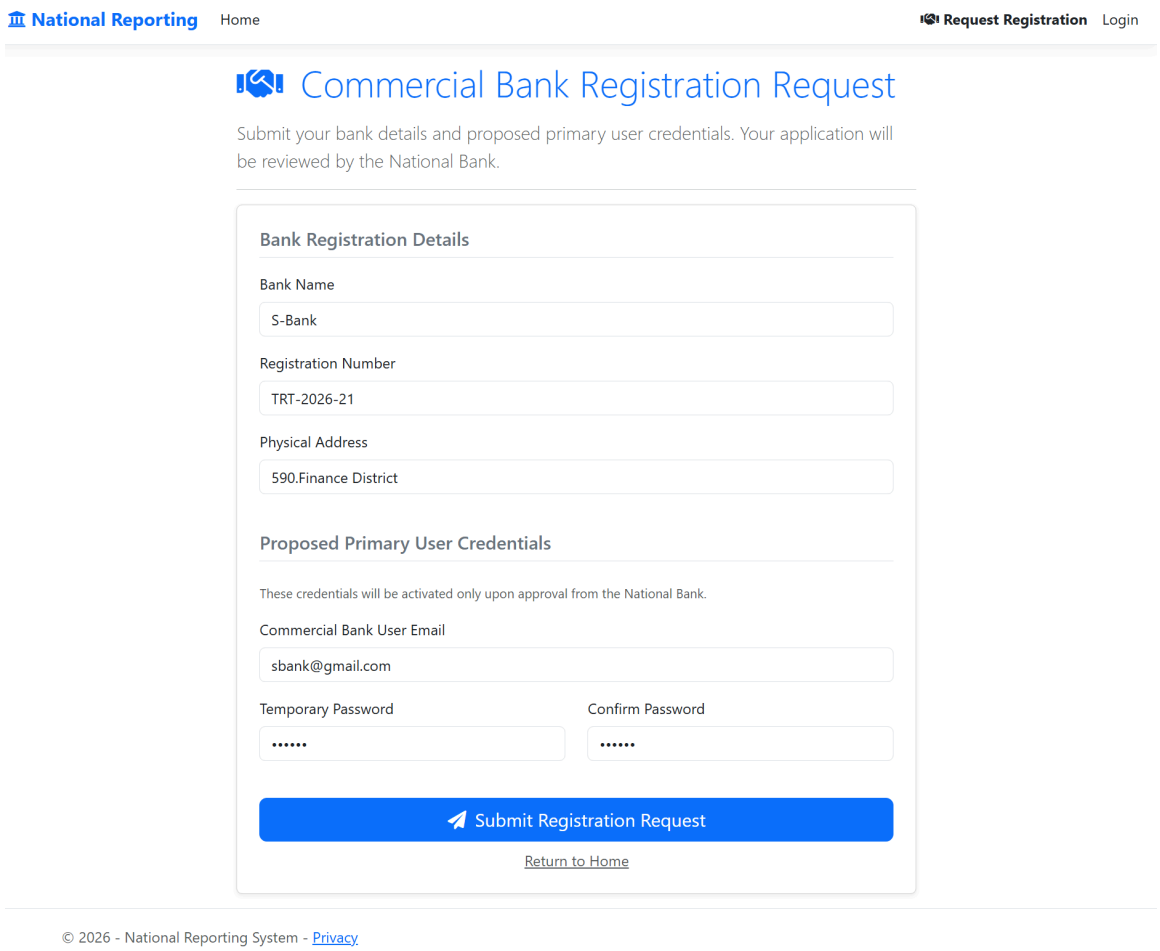


Figure 17: National Bank Reporting System Registration Request Interface

The Registration Request interface acts as the primary onboarding gateway for new institutions, enforcing a strict "verification-first" protocol. On this page, an unregistered entity must provide comprehensive institutional metadata, including the Bank Name, official Registration Number, and Physical Address, alongside proposed credentials for their primary user. From a security standpoint, the system does not immediately create an active user account upon submission; instead, the data is staged in the Banks table with a "Pending" status. This design prevents unauthorized access, as the "Proposed Primary User Credentials" remain inactive and the login functionality is blocked until a National Bank Administrator manually audits and approves the request within the administrative dashboard.

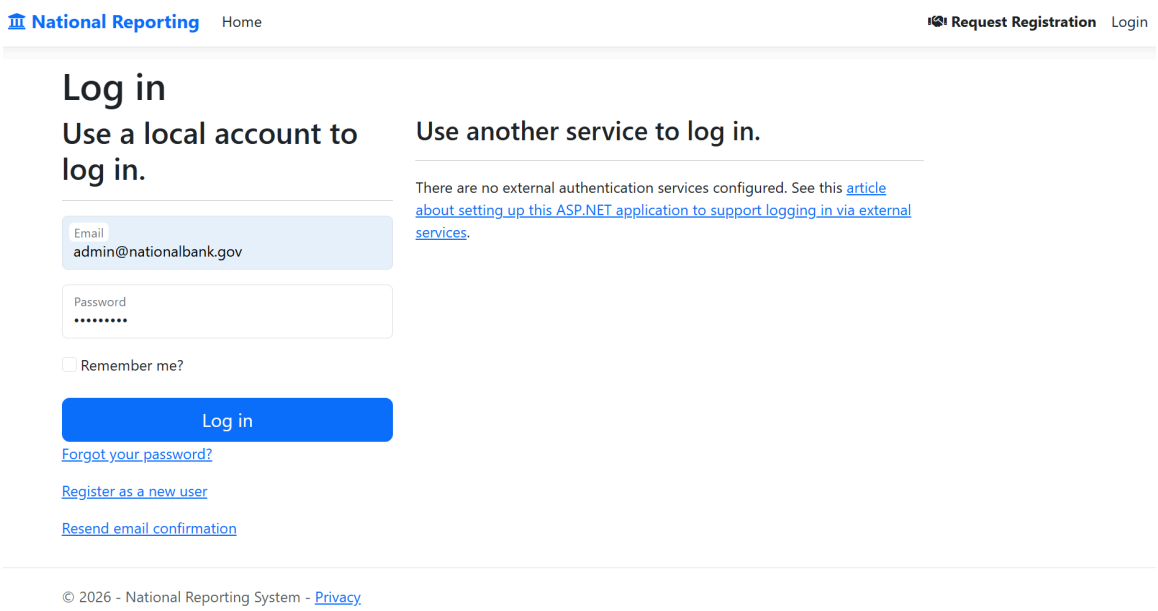


Figure 18: National Bank Reporting System Login Page

The Administrative Authentication interface serves as the primary secure entry point for the National Bank supervisory authority. To facilitate system testing and validation, a dedicated administrative account has been provisioned with the credentials *admin@nationalbank.gov* and the temporary placeholder password *Admin123!*. While this simplified configuration is utilized to streamline the development and demonstration phases, the production environment would enforce enterprise-grade security protocols, including complex password requirements and periodic mandatory credential rotation. Upon successful authentication, the system leverages ASP.NET Core Identity to verify the "NationalBankAdmin" role, granting the user full access to high-level oversight functions, institutional vetting, and financial audit dashboards.

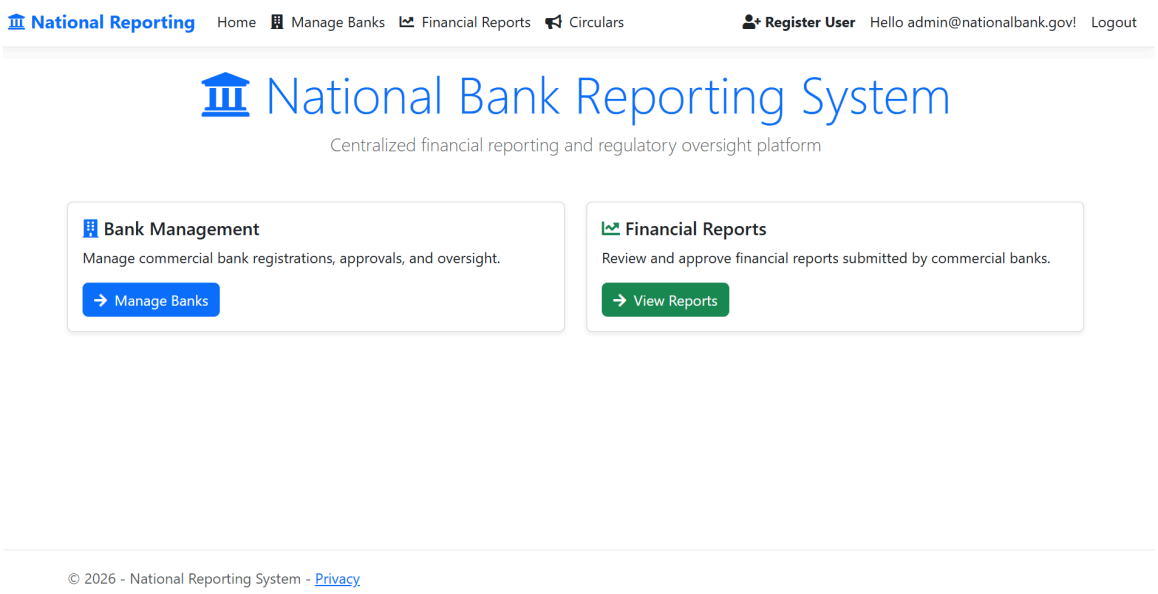


Figure 19: National Bank Reporting System Administrator Dashboard

The Administrator Dashboard serves as the central command center for the National Bank's supervisory activities. Upon successful authentication as an administrator, the interface dynamically updates to provide

exclusive access to critical oversight modules: Bank Management and Financial Reports. The "Bank Management" section allows the administrator to vet new institutional registration requests and manage existing commercial bank profiles, while "Financial Reports" provides a comprehensive view of all submissions awaiting regulatory review. Additionally, the navigation bar expands to include management links for Circulars and User Registration, ensuring the administrator has the tools necessary to maintain both regulatory communication and system security across the platform.

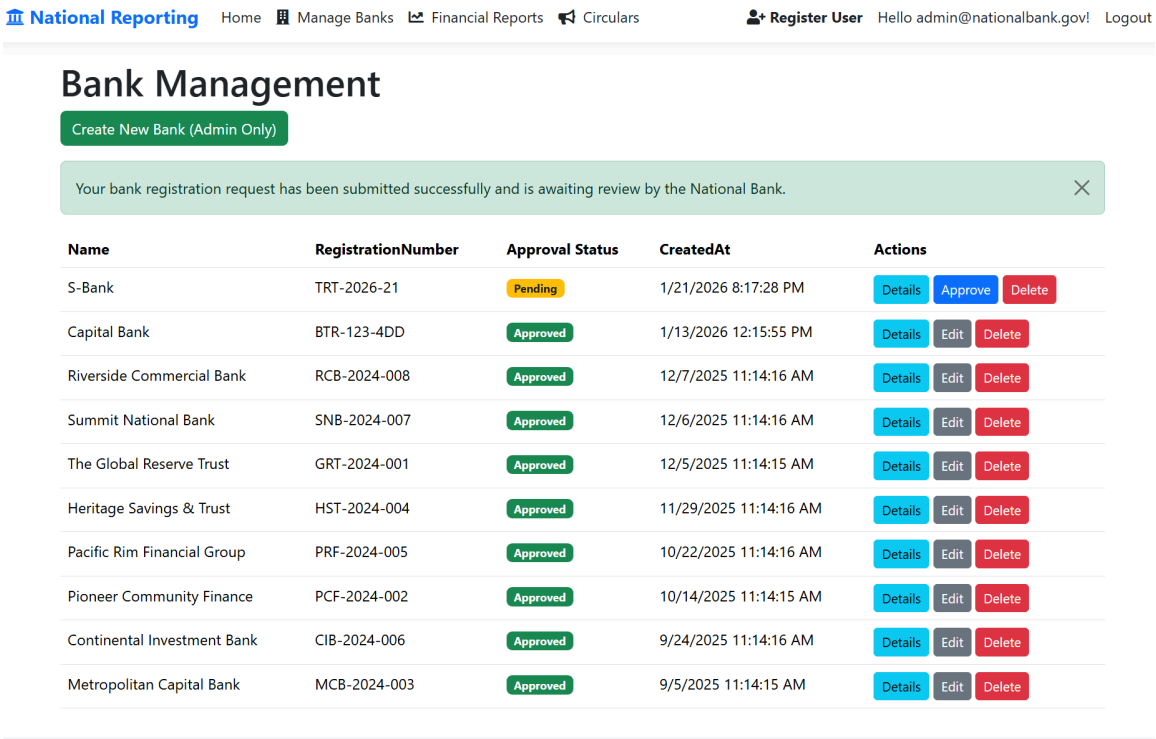


Figure 20: National Bank Reporting System Bank Managment Module

The Bank Management module provides the National Bank Administrator with a centralized interface to oversee the entire institutional ecosystem. This comprehensive view enables critical regulatory actions, including the ability to view details, edit, or delete existing commercial bank records. A key feature of this interface is the real-time status tracking of institutional onboarding; as illustrated, a newly submitted registration request appears with a "Pending" status, awaiting formal administrative audit. From this dashboard, the administrator can perform a manual verification of the bank's credentials and choose to Approve the request, granting the institution system access, or remove the entry entirely if it fails to meet regulatory standards. Additionally, the interface supports a "Direct Registration" path via the Create New Bank button, allowing the administrator to bypass the public request phase and immediately provision a verified institutional account.

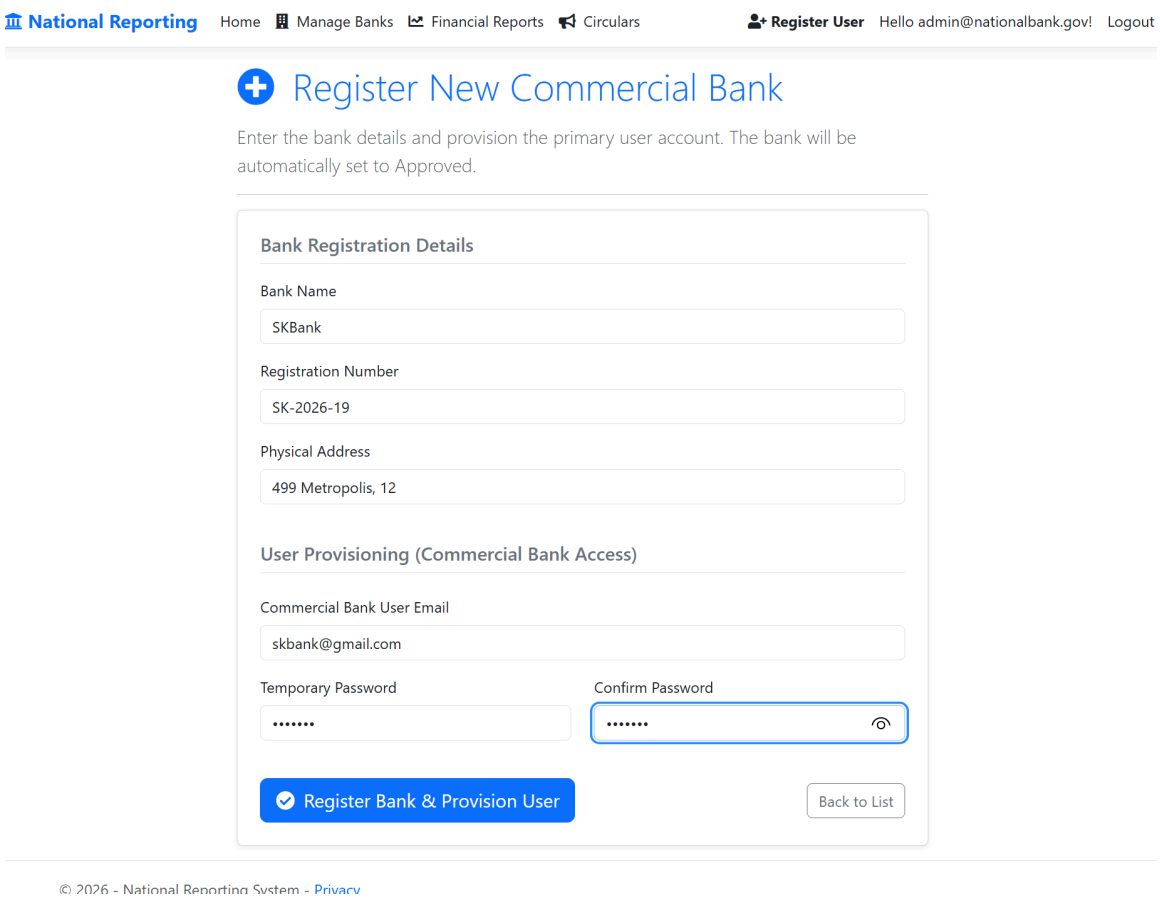


Figure 21: National Bank Reporting System Register Comercial Bank Inteface

The Register New Commercial Bank interface represents the system's "Direct Registration" workflow, reserved exclusively for the National Bank Administrator. Unlike the public request portal, this administrative tool allows for the simultaneous creation of a bank profile and the provisioning of its primary user account, with the institution's status automatically set to "Approved" upon submission. By entering the bank's legal metadata, such as the name "SKBank" and registration number "SK-2026-19", alongside the institutional email and temporary credentials, the Administrator ensures immediate system readiness for the commercial entity. This internal process bypasses the standard pending queue, providing a streamlined alternative for onboarding verified institutions while maintaining a centralized audit trail of all manually provisioned accounts.

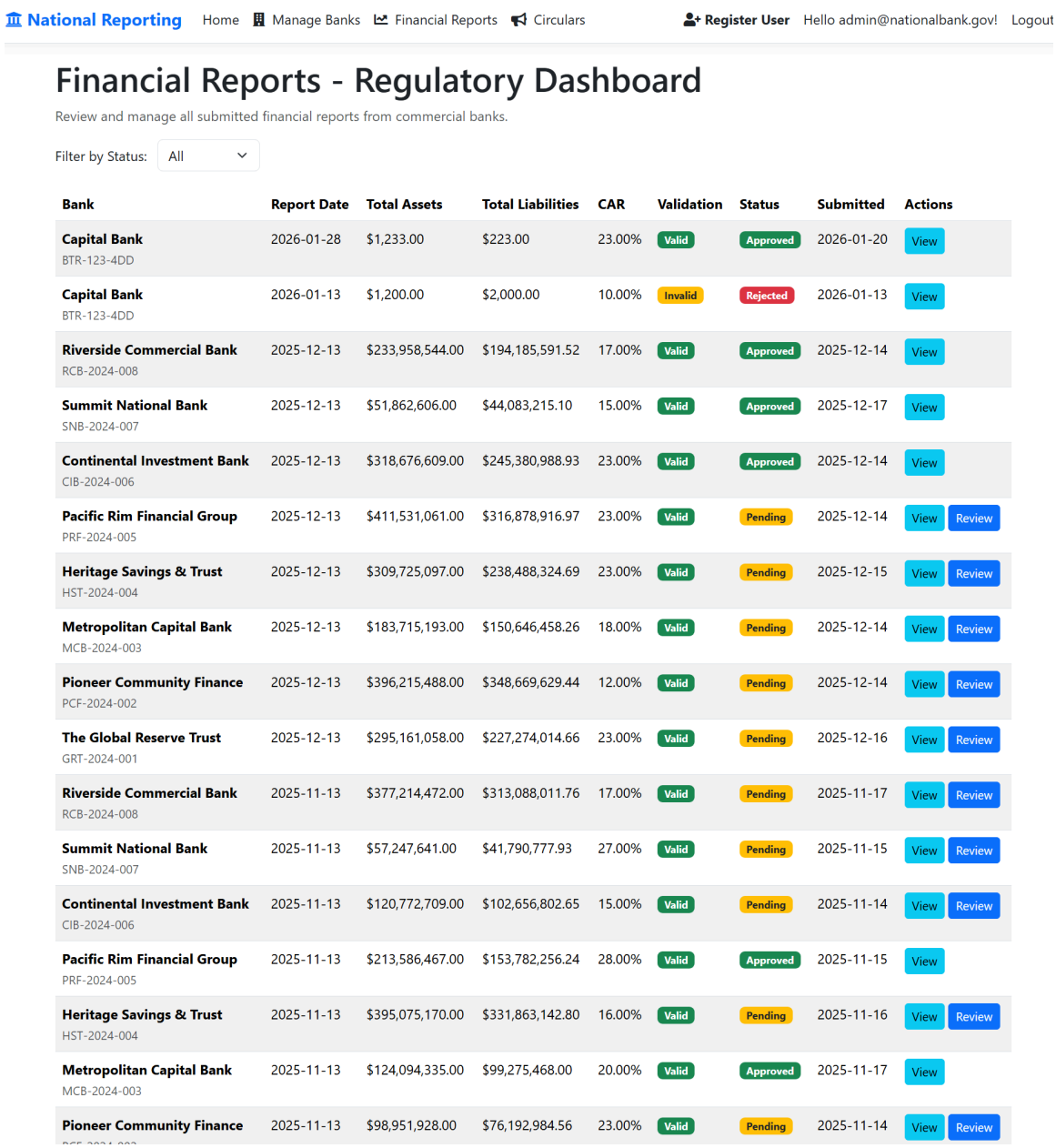


Figure 22: National Bank Reporting System Financial Report Regulation Dahsboard

The Financial Reports Regulatory Dashboard provides a centralized, data-driven view for the National Bank Administrator to audit institutional submissions. As shown in the interface, the system supports high-level filtering by status: All, Pending, Approved, or Rejected, allowing regulators to prioritize active tasks. Each entry displays critical financial metrics, including Total Assets, Total Liabilities, and the automatically calculated Capital Adequacy Ratio (CAR), which serves as an immediate health indicator. For "Pending" reports, the Administrator is granted exclusive Review capabilities to formally approve or reject the data based on its regulatory validity. Conversely, previously processed reports are locked for historical reference, with a View option available to ensure full transparency and a clear audit trail of all past regulatory decisions.

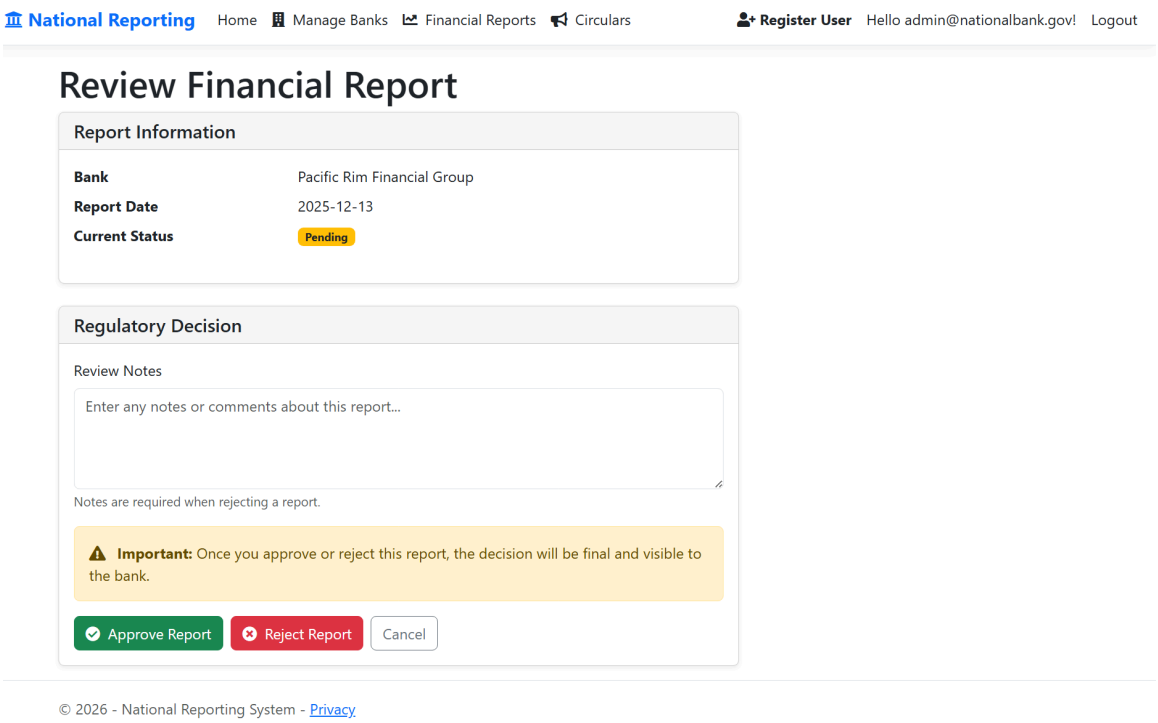


Figure 23: National Bank Reporting System Decision and Review Interface

The Regulatory Decision and Review interface provides the National Bank Administrator with the final layer of institutional oversight. When a "Pending" report is selected, the Administrator enters a dedicated review workspace where they can formally issue a Regulatory Decision, either Approve or Reject, based on the submitted financial data. A critical component of this workflow is the Review Notes field, which allows the Administrator to provide qualitative feedback, justify a rejection, or highlight specific areas for institutional improvement. Once the decision is saved, the report's status is permanently updated in the database, ensuring that the commercial bank receives immediate notification of the outcome while maintaining an immutable audit trail of the National Bank's supervisory actions.

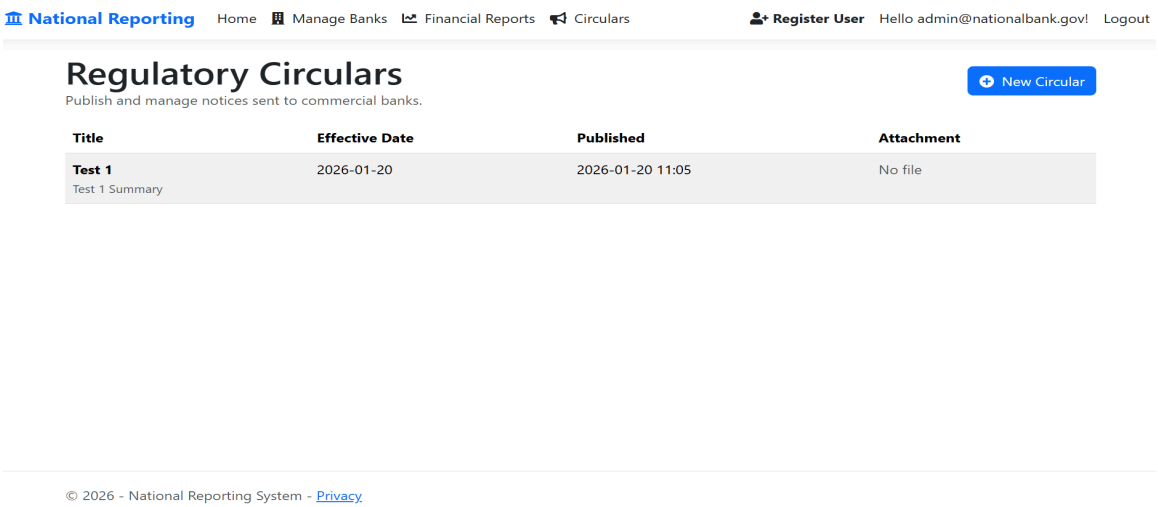


Figure 24: National Bank Reporting System Regulatory Circular Managment Interface

The Regulatory Circulars management interface serves as the official communication channel between the National Bank and the commercial financial sector. From this dashboard, the Administrator can publish and manage legal notices, policy updates, or reporting mandates that are broadcasted to all registered commercial banks. Each entry includes essential metadata, such as the Effective Date, Publication Timestamp, and an optional File Attachment for detailed documentation. By providing a centralized repository for these circulars, the system ensures that every commercial institution remains synchronized with the latest regulatory requirements, while the "New Circular" function allows for real-time dissemination of critical financial directives across the entire national network.

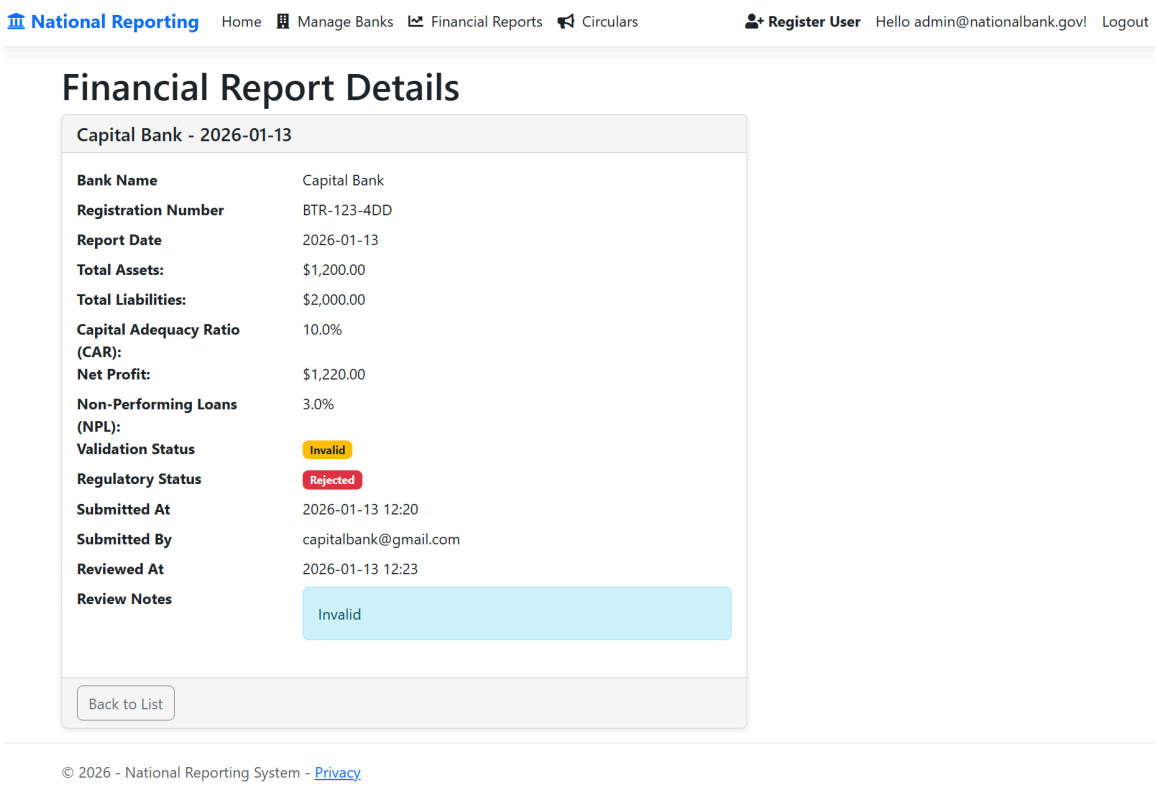


Figure 25: National Bank Reporting System Financial Report Details Interface

The Financial Report Details interface provides a comprehensive post-audit summary of a specific institutional submission. For the report shown, which belongs to Capital Bank, the system displays all captured financial metrics alongside a clear "Rejected" regulatory status and an "Invalid" validation status due to the institution's liabilities exceeding its assets. A critical feature of this view is the Review Notes section, where the Administrator's feedback—in this case, a simple "Invalid" notification, is permanently displayed to ensure the commercial bank understands the grounds for the regulatory decision. By documenting the exact submission and review timestamps, as well as the identities of both the submitter and the reviewer, the system maintains a complete and transparent lifecycle for every financial report processed by the National Bank.

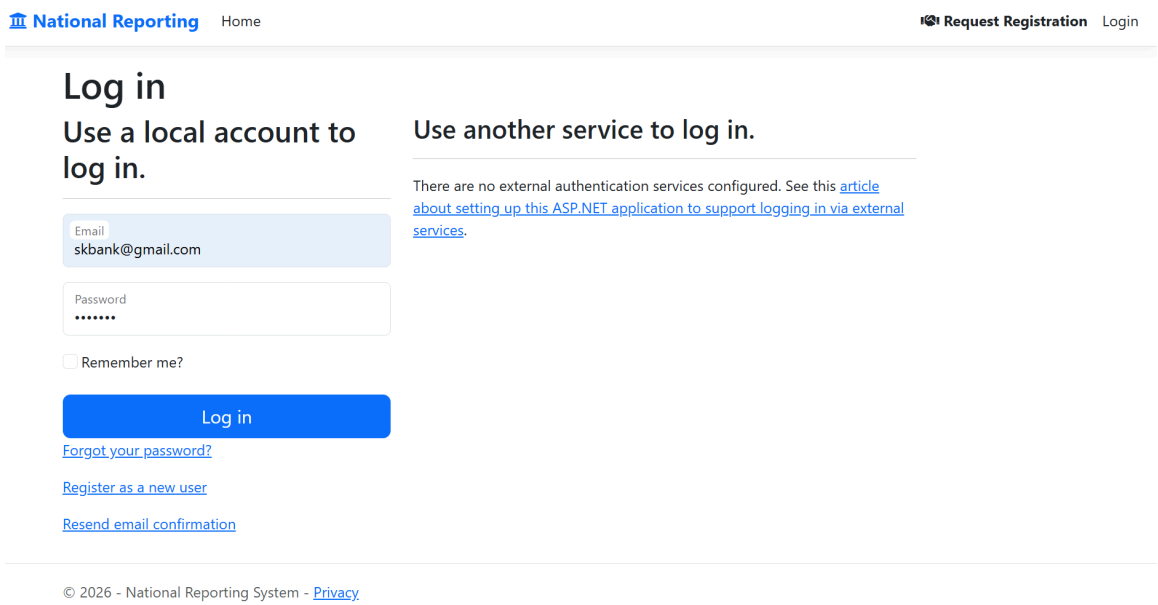


Figure 26: National Bank Reporting System Login page as a Commercial Bank

The Institutional Authentication phase follows the administrative provisioning of a commercial entity. Having been successfully registered by the National Bank Administrator, the bank representative can now utilize their assigned credentials, such as skbank@gmail.com, to access the secure reporting portal. This login process transitions the user from a public state to a verified Commercial Bank User role, as defined by the underlying ASP.NET Core Identity framework. Upon successful validation of these institutional credentials, the system redirects the user to a personalized dashboard tailored specifically for their bank's reporting obligations and history.

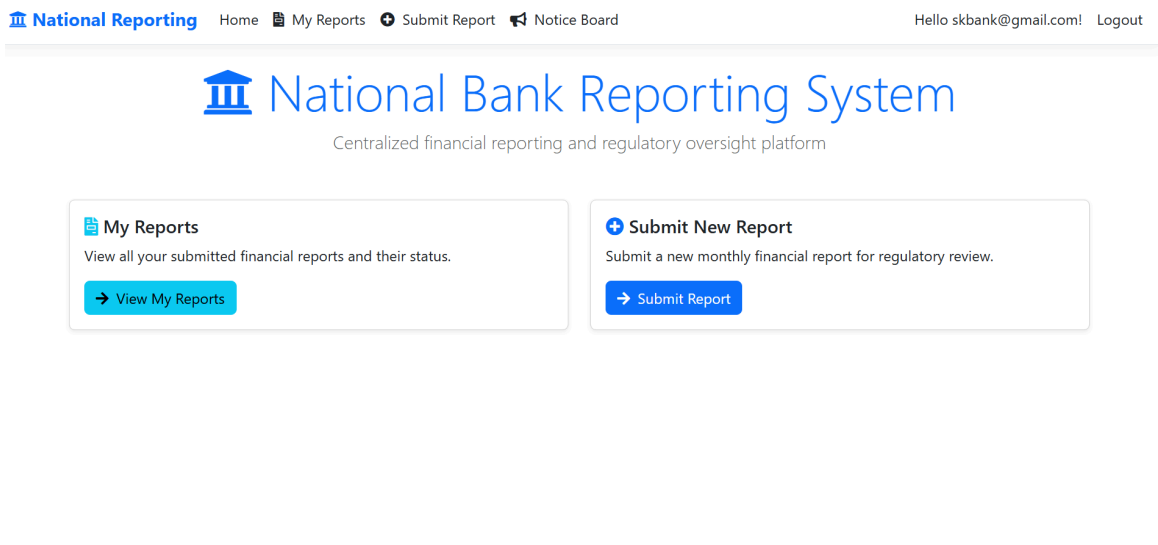


Figure 27: National Bank Reporting System Commercial Bank Dashboard

Upon successful authentication as a verified institutional representative, the system directs the user to the Commercial Bank Dashboard. This personalized workspace is specifically filtered to show only the data and actions relevant to the logged-in institution, such as "SKBank". The interface provides two primary

functional pathways: My Reports, which allows the user to monitor the historical status and regulatory feedback of all previous submissions, and Submit New Report, which initiates the standardized monthly data entry process. By dynamically updating the navigation bar to include institutional links like the Notice Board, the dashboard ensures the bank remains compliant with the latest National Bank circulars while maintaining a secure, role-based environment for its financial reporting obligations.

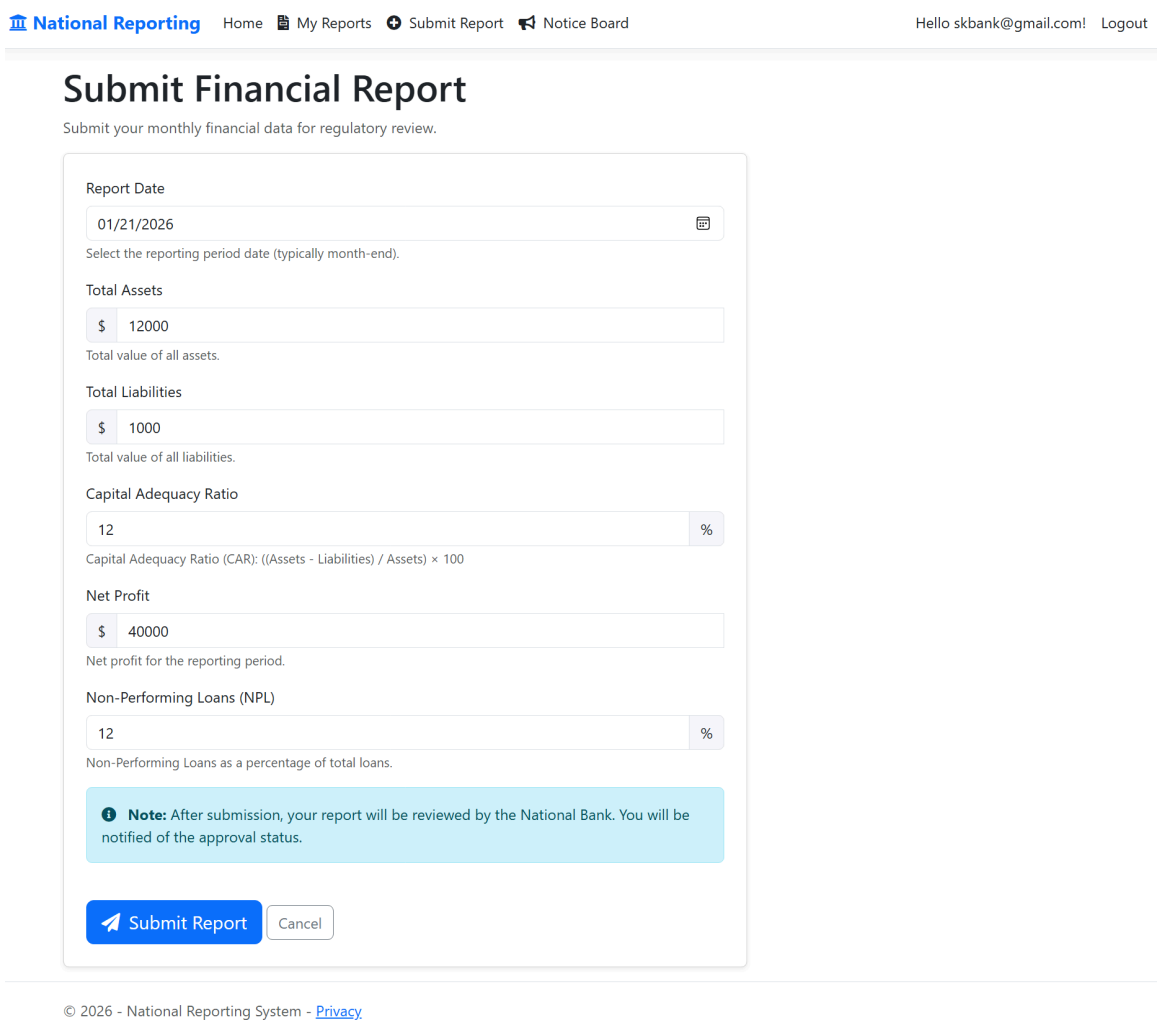


Figure 28: National Bank Reporting System Financial Report Submission Interface

The Submit Financial Report interface provides a standardized digital form for commercial banks to fulfill their periodic reporting requirements. This module streamlines the data entry process by requiring essential financial metrics, including Total Assets, Total Liabilities, and Net Profit. A critical technical component of this form is the automated calculation of the Capital Adequacy Ratio (CAR), which is derived using the formula: $CAR = ((Assets - Liabilities) / Assets) * 100$. Additionally, banks must report their Non-Performing Loans (NPL) as a percentage of total loans to provide a clear picture of institutional risk. Upon submission, the report is saved with a "Pending" status in the FinancialReports table, linked to the specific bank and submitting user, awaiting formal review and approval by the National Bank Administrator.

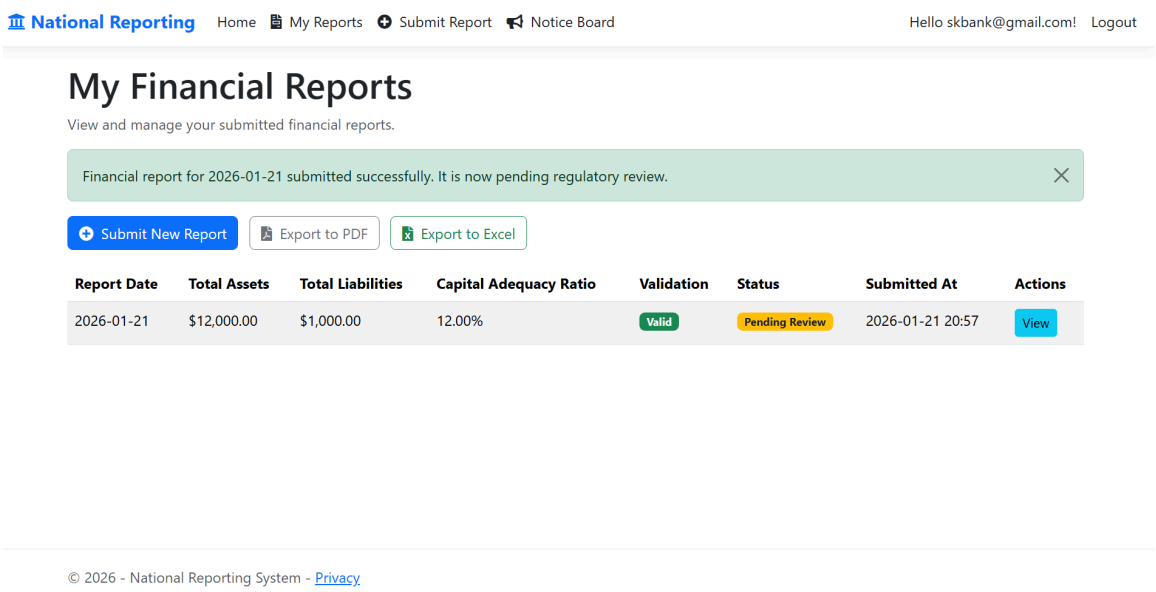


Figure 29: National Bank Reporting System Financial Report Interface

The My Financial Reports interface serves as the institutional ledger for the commercial bank, providing immediate confirmation of a successful submission. Upon completing the "Submit Financial Report" form, the user is automatically redirected to this view, where the new entry is listed with a "Pending Review" status, indicating it has entered the National Bank's regulatory queue. The dashboard presents a high-level summary of the submitted data, including the Report Date, Total Assets, and the Capital Adequacy Ratio (CAR), while also confirming the record's "Valid" internal validation status. To support institutional record-keeping and offline auditing, the system provides integrated Export to PDF and Export to Excel functionalities, allowing the commercial bank to maintain local copies of their regulatory filings as they await formal administrative approval.

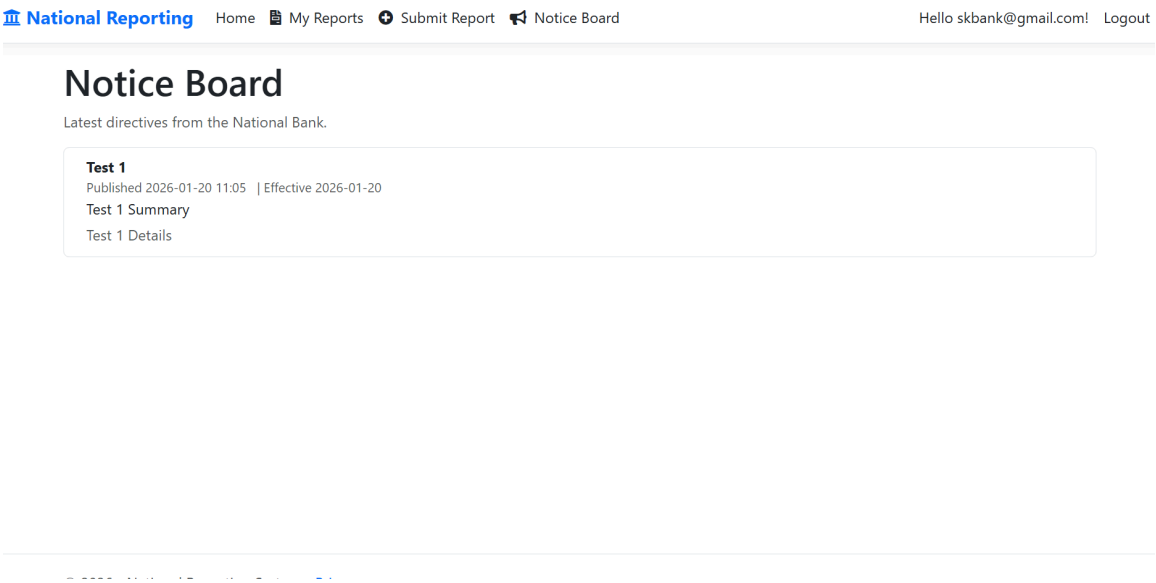


Figure 30: National Bank Reporting System Notice Board

The Notice Board serves as the institutional counterpart to the administrator's circular management module, providing a dedicated space for commercial banks to review official directives. Upon accessing this section, a commercial bank user, such as the representative for SKBank, is presented with the latest communications from the National Bank, including titles, summaries, and full details of each notice. As illustrated by the "Test 1" entry, the platform clearly displays both the Publication Date and the Effective Date to ensure institutional compliance with specified timelines. This centralized communication hub eliminates the fragmentation of email-based notices, ensuring that all commercial entities have immediate and permanent access to the regulatory framework governing their operations.

In conclusion, this integrated workflow demonstrates the successful transition from fragmented, manual communication to a centralized, role-based oversight system. By providing the National Bank with a dedicated interface to publish directives and the Commercial Bank with an immediate Notice Board, the system ensures regulatory synchronization across the financial sector.

Key achievements within this phase of the development include:

1. Elimination of Communication Gaps: Official mandates are no longer subject to the delays of email or physical document exchange.
2. Enhanced Compliance Tracking: Commercial banks can view effective dates and summaries in real-time to prioritize institutional responses.
3. System Integrity: The entire lifecycle—from the Administrator's creation of a circular to the institutional user's dashboard view—is maintained within a secure, authenticated environment, ensuring an immutable record of all regulatory communication.

This concludes the functional roadmap, proving that the National Bank Financial Reporting and Supervision System provides a robust, auditable, and transparent framework for modern banking oversight.

7 Discussion

The implemented system effectively addresses the challenges identified in the problem statement. Centralized data storage eliminates fragmented communication and ensures consistency, while structured forms and validation rules enforce standardized reporting formats. Role-based authorization guarantees that only authorized users can perform sensitive actions, and audit logging provides transparency and accountability. As a result, the system improves efficiency, reliability, and trust in the financial reporting and supervision process.

8 Conclusion and Future Work

This project demonstrates the successful design and implementation of a National Bank Financial Reporting and Supervision System using modern .NET technologies. The system models real-world regulatory workflows while maintaining security, scalability, and maintainability. It also serves as a strong educational example of applying ASP.NET Core MVC, Entity Framework Core, and Identity in a practical domain.

Despite its effectiveness, the system has certain limitations, including the absence of advanced analytics and real-time dashboards. Future work may include integrating data visualization tools, automated report generation, extended regulatory roles, and deployment to a cloud-based infrastructure.